Literature status: DSS inclusions between Nakano function spaces

Automated Math Discovery Run `fa_banach_001`
Model: GPT5.6

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Status: literature already answered. Hernández–Ruiz–Sanchiz (2024) give the complete characterization asked for by Ruiz–Sánchez (2014), for both finite and infinite atomless measure spaces.

Source question

Ruiz and Sánchez close their paper with three research directions. The middle one states: "a characterization of disjointly strictly singular inclusion operators between Nakano function spaces is still unknown" [1, p. 11, Open questions]. Here Nakano function spaces are the variable-exponent Lebesgue spaces $L^{p(\cdot)}$.

The exact source is arXiv:1401.5906v5, PDF page 11. The local copy is `source_paper.pdf`.

Decisive later classification

Hernández, Ruiz, and Sanchiz state in their abstract and introduction that they characterize DSS inclusions between variable Lebesgue spaces and obtain complete characterizations [2]. On an atomless finite measure space, their Theorem 4.8 (PDF pages 15–16) says that, when $q(\cdot) < p(\cdot)$ almost everywhere, the following are equivalent:

$$\int_0^{\mu(\Omega)} a^{\left(\frac{pq}{p-q}\right)^*(x)}\,dx < \infty \quad \text{for every } a > 1,$$
$$L^{p(\cdot)}(\mu) \hookrightarrow L^{q(\cdot)}(\mu) \quad \text{is DSS.}$$

It also proves equivalence with L-weak compactness, M-weak compactness, a normalized-indicator test, and an explicit rearrangement limit. If $p = q$ on a positive-measure set, the inclusion is trivially non-DSS, so the criterion covers every inclusion on the finite atomless space.

For a non-atomic infinite measure space, Theorem 6.3 (PDF page 21) proves the complementary classification: whenever $L^{p(\cdot)}(\mu) \hookrightarrow L^{q(\cdot)}(\mu)$ exists, it is not DSS.

Identification and scope

This is the exact object named in the 2014 question, with no change of terminology: Nakano function spaces are $L^{p(\cdot)}$, and the operator is the canonical inclusion. The 2024 paper explicitly claims complete characterizations; moreover, César Ruiz is an author of both sources. Accordingly this is a direct later literature answer, not a proof produced by this run.

The source's neighboring suggestions about algebrability and about broader spaceability phenomena in Nakano spaces are separate questions and are not claimed to be resolved here.

References

[1] C. Ruiz and V. M. Sánchez, *Nonlinear subsets of function spaces and spaceability*, arXiv:1401.5906v5 (2014). `https://arxiv.org/abs/1401.5906`

[2] F. L. Hernández, C. Ruiz, and M. Sanchiz, *Disjointly strictly singular inclusions between variable Lebesgue spaces*, arXiv:2406.14175 (2024). `https://arxiv.org/abs/2406.14175`